CHAPTER FOUR

Interception of Subsurface Lateral Flow Through Enhanced Vertical Preferential Flow in an Agroforestry System Observed Using Dye-Tracing and Rainfall Simulation Experiments

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Contents

Abstract

Surface soil hydrology is a major control on the terrestrial water cycle in Earth's Critical Zone (CZ). Partitioning of vertical preferential flow and subsurface lateral flow is commonly attributed to the heterogeneity of slope, soil profile horizon, and soil structure, but the influences of land-use types are largely unknown. Agroforestry systems (AF) can

Advances in Agronomy, Volume 142
ISSN 0065-2113
http://dx.doi.org/10.1016/bs.agron.2016.10.014

intercept subsurface lateral flow for reducing nitrogen losses to drainage water and for alleviating secondary salinity in central China and southeast Australia. These effects have been attributed to enhanced evapotranspiration and canopy interception in the agroforestry systems compared to monocropping systems (MC). Here, we show the differences in lateral and vertical soil hydrological pathways between AF and MC with dye-tracing experiments before and during a simulated rainfall event. Before the rainfall, the vertical and horizontal dye-staining patterns demonstrated that preferential flow occurred through isolated macropores with fine tree roots in AF and through connected cracks in MC. The dye coverage area and depth indicated greater vertical preferential flow in AF than in MC. During the 2-h rainfall event, the dye-staining area at different depths indicated that the preferential flow contributed to greater near-surface lateral flow in MC than in AF. The changes in the hydrological pathways were attributed to deep roots and no physical barrier from plough pan in AF and the presence of the plough pan in MC. These results suggest that land use has strong water partitioning effects not only above ground but also in the subsurface, and that understanding the landscape hydrology in the Earth's Critical Zone required quantification of the considering coupled pedological and biological processes.

1. INTRODUCTION

Earth's Critical Zone (CZ), extending from the top of the vegetative canopy down to the deepest fresh groundwater circulation, involves the flows of energy, water, carbon, and nutrients, thereby sustaining the natural resources that upon which mankind depends (Brantley et al., 2007; Lin, 2010). Vertical preferential flow and subsurface lateral flow are not only common surface soil hydrological processes (Wang et al., 2011; Zhang et al., 2011) but are also major physical controls on major environmental and sustainability challenges (i.e., drought, flood, soil erosion, water pollution). These challenges facing mankind today involve important linkages with vertical preferential flow and subsurface lateral flow in Earth's Critical Zone (Lin, 2010; Zhang et al., 2011). These water flow pathways transport water and pollutants vertically to deep soil as recharge to groundwater, and laterally along slopes into low-lying water bodies (Schaik, 2010; Wang et al., 2011; Zhang et al., 2011). Vertical preferential flow and subsurface lateral flow usually interact with each other due to soil heterogeneity in soil physical properties and process rates (Alletto and Coquet, 2009; Schaik, 2010; Zhang et al., 2015a). Therefore, the hydrological interactions may drastically change under various land uses that are a major source of soil heterogeneity in Earth's Critical Zone. However, the effects of land use on the hydrological interactions are still poorly understood, presenting a major barrier to solutions to these environmental and sustainability challenges.

Vertical preferential flow generally occurs when infiltrating water flows through soil macropores and bypasses most of soil matrix (Schaik, 2010). Subsurface lateral flow is often gravity-induced along the slope when sufficient water is impeded over a less permeable soil layer or bedrock within a soil profile (Shaw et al., 2001). Land use, subjected to natural and man-induced processes, is a source of soil heterogeneity (Alletto and Coquet, 2009; Schaik, 2010; Zhang et al., 2015b) that may influence the spatial variability in soil water flow (Flury et al., 1994; Forrer et al., 2000). Compaction of subsoil by using heavy machinery (Horn and Smucker, 2005) and the presence of plough pan beneath surface-cultivated soil (Heppell et al., 2000) can create a less permeable layer to reinforce infiltration resistance, which then induces subsurface lateral flow (Shaw et al., 2001). Macropores created by tillage (Zhang et al., 2015b), deeper penetration of plant roots (Schaik, 2010; Zhang et al., 2015a), and earthworm's activities (Trojan and Linden, 1992) can trigger vertical preferential flow in soils. Part of the flow variability can become negligible through flow mixing in the different directions (i.e., vertical and lateral). However, if soil heterogeneity is large and the flow mixing is limited, the land use will result in significant variability in vertical preferential flow and subsurface lateral flow (Allaire et al., 2009). Until now, as a source of soil heterogeneity, the connection between land-use practices and vertical preferential flow and subsurface lateral flow processes is unclear.

Knowledge of land use and subsurface lateral flow has been widely applied at the practical level for addressing the environmental and sustainability issues in Earth's Critical Zone. For instance, agroforestry systems were proposed in southeast Australia to alleviate secondary salinity and in central China to reduce nitrogen losses from fields in drainage water by intercepting subsurface lateral flow and nitrate transport (Dunin, 2002; Wang et al., 2011). However, the function of agroforestry systems to intercept subsurface lateral flow has been largely attributed to the enhanced evapotranspiration and canopy interception (Dunin, 2002; Wang et al., 2011). Preferential flow in agroforestry system may occur due to increased saturated conductivity (Seobi et al., 2005) and more biopores due to decayed roots (Schaik, 2010). Wang et al. (2011) found that more soil water can be retained in the soil profile, and reduced subsurface lateral flow can be generated, during rainstorm in an agroforestry system compared with a monocropping system. These studies implied that agroforestry systems may modify soil hydrological pathways to intercept subsurface lateral flow along hillslopes.

Dye-tracing techniques allow soil hydrological pathways to be both visualized and quantified at different physical scales of observation (Edwin et al., 2009; Forrer et al., 2000; Zehe and Flühler, 2001). Also, the interactions between morphological observations and soil hydrological process can be characterized when dye-tracing is used in conjunction with image analysis (Morris and Mooney, 2004; Sander and Horst, 2007). Until recently, most dye-tracing experiments were carried out under ponding condition to determine vertical hydrological pathway, while fewer experiments have been performed under rainfall conditions to study lateral flow pathways. Therefore, with dye-tracing experiments before and during a simulated rainfall, the objectives of this study were to characterize the vertical and lateral soil hydrological pathways along the slope and to determine the effects of an agroforestry system (AF) on the interaction between lateral and vertical soil water flows compared with a monocropping system (MC).

2. MATERIALS AND METHODS

2.1 Experimental Site

The study was carried out in a small agricultural catchment (Zepp et al., 2005), about 5 km from the Ecological Experiment Station of Red Soil, Chinese Academy of Sciences (EESRS, CAS) in Yingtan, China. The research area had a typical subtropical climate with a mean annual temperature of 17.7°C, a maximum daily temperature of around 40°C in summer, and an annual rainfall of 1786 mm. The elevation at the catchment ranges from 41 to 55 m above the sea level and the soil depth ranges from 0.4 to 2.0 m. The upland soils were developed from Quaternary clay and are classified as loam clay Ultisols according to the Soil Taxonomy (Soil Survey Staff, 2010).

Because this study aimed to differentiate the agricultural land-use effects on lateral and vertical soil hydrological pathways, two cropping systems were compared on the same slope. One was agroforestry system (AF) with citrus trees (*Citrus reticulata*) intercropped with peanut crops (*Arachis hypogaea* L.) in a downslope alley and another was monocropping of peanut system (MC) located in parallel to AF. The trees and crops were planted in rows down the slope, which was about 6% in gradient and 130–150 m long. The 20-year-old citrus trees were planted in rows of 4 m × 4 m along the slope and intercropped for 10 years. The citrus trees were about 3 m high, with the roots penetrating to 1.1 m soil depth and the leaf area index ranging from 4.4 to 5.5 during the growth period. Peanut crops were sown at a spacing of 0.20 m × 0.30 m in the middle of April and harvested in the middle of August in both systems. The peanut crops in AF were planted in 1.5-m-wide strips between the citrus trees, and all the field management practices were identical for the peanut crops.

One soil profile under each cropping system was excavated and characterized following soil survey protocols in the field (FAO/ISRIC/ISSS, 1998). Soil samples were taken by soil horizon for physical and chemical analysis. Soil particle size distribution was measured with the pipette method (Lu, 1999). Five soil cores taken from each soil horizon were used to measure soil bulk density. Soil organic carbon was measured after wet digestion with $K_2Cr_2O_7$–H_2SO_4 (Kalembasa and Jenkinson, 1973). The selected soil properties are presented in Table 1.

2.2 Dye-Tracing Experiment Without Rainfall Simulation

The dye-tracing experiment without rainfall simulation was conducted at the same slope position under both land uses following the method of Sander and Horst (2007). Briefly, two 1.0 m × 1.0 m steel frames as replicates were inserted to the soil depth of 0.10 m in the two systems. The frames were placed 0.5 m away from two tree rows in the untilled alley in AF and in the tilled area in MC. The soil around the frame walls was compacted to prevent bypass flow, and a plastic sheet was used to cover the frames before adding Brilliant blue (CAS number: 6104-59-2) solution (5 g L^{-1}). Fifty-liter dye solution was carefully poured over the plastic sheet, forming a 50 mm ponding water head. The plastic sheet was rapidly removed from one side of the frame to start infiltration, and then the frame was covered with plastic sheet to prevent evaporation. Twenty-four hours after infiltration, a 1.0 m × 1.0 m access pit was excavated along one side of the frame after removing the frame. The dyed area within the frame was separated into two parts from the middle line, for analysis of vertical and horizontal profiles. For vertical profile analysis, photos were taken within a rectangular frame (1.0 m × 1.0 m) as a spatial reference. The vertical profile was photographed and moved at an internal distance of 0.1 m from the

Table 1 Selected Chemical and Physical Properties of Soil Profiles in the Agroforestry System (AF) Consisting of Peanut and Citrus and Monocropping System (MC)

Land Use	Horizon[a]	Depth (m)	Bulk Density ($Mg\ m^{-3}$)	Soil Organic Carbon ($g\ kg^{-1}$)	Particle Size Distribution (%)			Comments
					2–0.05 mm	0.05–0.002 mm	<0.002 mm	
Agroforestry	A_p	0–0.2	1.36	10.85	42.49	24.65	32.86	Prismatic/blocky; surface compacted, rich in fine root, biopores formed from decayed root
	B_1	0.2–0.5	1.43	2.705	38.19	25.96	35.85	Prismatic/blocky; rich in root; some biopores formed from decayed root
	B_2	0.5–0.9	1.51	2.59	36.45	25.75	37.80	Granular/crumbs; root distributed
	B_{3t}	0.9–1.5	1.60	—	34.04	26.79	39.17	Coherent and single grain; rare root
Monocropping	A_p	0–0.2	1.35	11.22	41.81	25.80	32.39	Prismatic/blocky, surface crust; loose; rich in fine root, biopores formed from decayed root
	B_1	0.2–0.5	1.47	2.9	39.56	26.18	34.26	Prismatic/blocky; some biopores formed from decayed root
	B_2	0.5–0.9	1.51	2.44	35.69	26.71	37.60	Granular/crumbs
	B_{3t}	0.9–1.5	1.60	—	32.07	26.93	41.00	Coherent and single grain

[a]t, accumulation of silicate clay.

edge to the middle of the metal frame, and five vertical profiles were photographed in each plot. For the horizontal profile analysis, horizontal photos were taken within a rectangular frame (0.5 m $\times$ 1.0 m) as a spatial reference, and the surface soil was excavated at an interval of 0.05 m downward. Before photographing, the soil surfaces were brushed to remove the crumbs carefully using a knife. The photos were taken using a digital camera (Olympus E510, 3264 by 2448 pixels) under a black sheet to avoid strong contrasts by unequal sunlight exposure of the photographs. Altogether, 54 horizontal and 20 vertical profiles were photographed in both systems.

2.3 Dye-Tracing Experiment With Rainfall Simulation

Dye-tracing experiment with rainfall simulation was conducted in confined erosion plots (3.5 m $\times$ 14.0 m) at the downslope position in both systems during the period from 15th to 30th, April, when is the start month of rainy season in study region. The plots were plowed to 0.15 m to prepare peanut sowing. A set of tensiometers were installed in the middle of each plot at the depths of 0.10, 0.20, 0.30, 0.40, 0.60, 0.85, 1.15, and 1.50 m to monitor soil matric potential during rainfall simulation. A steel ring (0.3 m diameter $\times$ 0.4 m in height) was inserted to the soil depth of 0.10 m at the same slope position in the erosion plots, where the soil conditions (approximately -80 hPa at the 0.1 m depth and -20 hPa at the 1.50 m) were similar to those as described for the dye experiment under ponding infiltration. The ring was used to conduct dye-tracing experiment during rainfall simulation. Three point fifty three liter Brilliant blue solution (5 g L^{-1}) was poured into the ring carefully before the rainfall simulation, resulting in a 50 mm ponding water head. The rainfall simulation lasted for 2 h. During the rainfall simulation, the steel ring was covered with a plastic sheet to prevent rainfall falling into the ring. The rainfall intensity was designed differently for the two systems to ensure the same infiltration, considering the difference in through-fall in AF. Ten plastic cups of 0.15 m in diameter and 0.2 m in height were randomly placed in open area in both erosion plots to calibrate rainfall intensity. And another ten cups were placed under the tree canopy in AF to weigh through-fall. Overland flow was measured by a tipping bucket system that was installed at the lower end of each plot. The number of bucket tips was recorded using an event data logger (Onset Computer Corporation, USA) to calculate the overland flow volume. The recorded rainfall in 2 h was 97.6 mm in AF and 68.0 mm in MC. The overland flow was 46.7 mm in AF and 29.7 mm in MC; the through-flow was 11.7 mm in AF. Thus, the amount of infiltration into soil during the rainfall simulation was 39.2 mm in AF and 38.3 mm in MC.

When the soil matric potential decreased back to the initial values before rainfall simulation (about 23 h after the cease of rainfall simulation) in both systems, the steel ring was removed to take horizontal photos within a 1.5×1.5-m frame that had the ring as the center for 0.1 and 0.2 m depths following the similar procedures as described above. Briefly, the soil surface was brushed and crumbs were removed after the soils were excavated. The distance of dye movement upslope and downslope from the center for the ring was measured using a ruler. The horizontal soil surfaces were photographed, together with a wooden spatial reference frame ($1.5 \text{ m} \times 1.5 \text{ m}$) and four Kodak color and gray scales onto the each side of the frame, following Forrer et al. (2000) and Kasteel et al. (2007). A black cover sheet was used to avoid strong contrasts during photographing.

During the excavation, 10 bulk soil cores (100 cm^3) were collected from the top soil apart from the ring in each plot. In the laboratory, the soil cores were saturated for 6 days in solution with different dye concentration: 0.1, 0.5, 1, 2, 4, 6, 20, 40, 80, and 150 $g\,L^{-1}$. Thereafter, the soil cores were freely drained at 25°C for 24 h, brushed with a knife on the surface, and used to take photographs using the same camera as for the field observations, in order to establish quantitative calibration between dye concentration and soil color intensity captured in the digital images (Alaoui and Goetz, 2008).

2.4 Image Analysis

The images obtained from the dye-tracing experiment without rainfall simulation were orthogonal with WGEO software (WASY GmbH, 2006) by transforming the four corners of photographed reference frames from internal raster to real coordinates. The undyed parts of the orthogonal images were removed with the Photoshop software (Photoshop 7.0). The images were transformed into binary format to calculate dye coverage in ImageJ software (National Institute of Mental Health, USA). To quantify the variability of soil water flow, the concept of actual penetration depth (H_a) was employed (Wang et al., 2006). An orthogonal three-dimensional coordinate system (X, Y, Z) was set into the soil columns for the analysis of horizontal dye-staining pattern (0.5 m × 1.0 m), with an X axis along the length direction (1.0 m), a Y axis along the width direction (0.5 m), a Z axis along the depth, and an origin (0,0,0) located at a corner of the soil surface. Each horizontal profile was split into 100 segments along the X axis direction and 50 segments along the Y axis direction; hence, the entire horizontal profile was separated into 5000 nodes, and each node covered an area of 0.01 m × 0.01 m. The actual penetration depth was calculated from Eq. (1):

$$H_a(x, y) = \sum_{i=1}^{k} \alpha(x, y, z)\Delta h_i \tag{1}$$

where, $H_a(x, y)$ is the actual penetration depth of the plot; $\alpha(x, y, z)$ is the coordinate of α, when the soil was dyed at the coordinate of (x, y, z), then $\alpha(x, y, z) = 1$, if not, then $\alpha(x, y, z) = 0$; Δh_i is the distance of the two adjacent nodes in vertical direction (Z axis direction).

The images obtained from the dye-tracing experiment under rainfall simulation condition were analyzed with a rigorous and repeatable method on the basis of calibrated dye concentrations. This method was employed in many other studies (Forrer et al., 2000; Kasteel et al., 2007). The first step was to orthogonalize the images with WGEO software. The second step was to adjust the color of images to warrant the comparability among images by applying a background subtraction technique through which the color and illumination were corrected by using Kodak gray scale applied to the each color channel (R, G, and B values). The third step was to determine the relationship between the dye concentrations and their associated mean values of every color channel ($\bar{R}$, $\bar{G}$, and $\bar{B}$ values) on the images by fitting a second-order polynomial using the standardized mean variables $\bar{R}$, $\bar{G}$, and $\bar{B}$ as independent variables, based on the correlation between dye concentrations and the value of color channels obtained in the laboratory. The fourth step was applied with the determined second-order polynomial function and the R, G, and B values fitted for each pixel in the processed horizontal images, for estimating the dye concentrations. To separate dyed and undyed soil in the images, the threshold concentrations were set between the highest measured concentration in the nonstained samples and the lowest measured concentration in the stained samples (Kasteel et al., 2007).

During the rainfall simulation, subsurface lateral flow can cause dye movement out of the surface area of the ring containing dye solution.

Assuming that preferential flow was the only flow through macropores in soil and that no chemical and physical reactions occurred between the soil matrix and dye solution, the area with dye concentration of >5 g L^{-1} during the rainfall simulation indicated that preferential flow occurred in the area. Therefore, the preferential flow contribution to subsurface lateral flow (C_p) was calculated based on Eqs. (2) and (3):

$$C_P = \frac{D_{md} - D_{ad}}{D_{md}} \times 100\% \tag{2}$$

$$D_{ad} = \sqrt{\frac{A_{stain}}{\pi}} \tag{3}$$

where D_{md} is the maximum distance of dye movement from the central point of the steel ring; d_{ring} is the radius of the steel ring (0.15 m); and D_{ad} is the actual distance of dye movement. It is assumed that when no subsurface lateral flow occurs, the wetness front of dye solution is uniform in the soil and the dye-staining area should be a circulate section in the horizontal images. Then, D_{ad} is the radius of the circle, which is calculated based on circle area (A_{stain}).

3. RESULTS

3.1 Vertical and Horizontal Dye-Staining Patterns Without Rainfall

Geometrically corrected photos of the vertical soil profiles are shown in Fig. 1. The fingering pattern of dye-staining fronts showed a high spatial density of vertical preferential flow in both systems. Comparing the two systems, the dye coverage was significantly larger in AF than in MC ($p < 0.05$) (Table 2). With the increasing soil depths, the dye coverage decreased evenly in AF, while it showed an abruptly decrease at the 0.15–0.17 m depth in MC, suggesting the presence of plough plan in MC. The dye-staining paths were more spatially related in MC than in AF, in which most of the dye staining was concentrated in the surface-ploughed layer and the preferential flow pattern started below the plough pan in MC. The photos taken horizontally with the metal frame demonstrated the dye-staining pattern at the 0.10 m soil depth (Fig. 2). The dye-stained flow paths were well connected along crack networks in the both systems, but the stained area along the cracks was wider and more extensive in MC than in AF. In addition, the

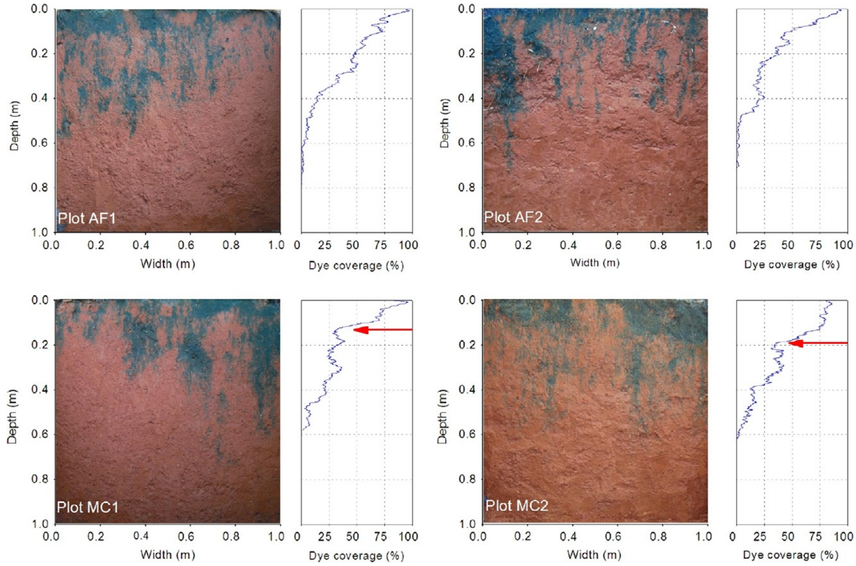

Fig. 1 Typical photos showing vertical dying patterns in the soil profiles under agroforestry system (Plot AF1 and AF2) and monocropping system (Plot MC1 and MC2) under the ponding infiltration without rainfall simulation. Dye coverage by soil depth was presented on the right hand side of each of the photos.

Table 2 The Dye Coverage and Plough Pan Depths Without Rainfall Simulation in the Vertical Images of Agroforestry (AF) and Monocropping System (MC)

Plot	Image Number (n)	Dye Coverage (%)		Plough Pan Depth (m)	
		Mean	SD	Mean	SD
Agroforestry AF1	5	15.1	1.6	—	—
Agroforestry AF2	5	15.9	1.6	—	—
Monocropping MC1	5	12.2	2.0	0.14	0.04
Monocropping MC2	5	13.0	1.5	0.17	0.04

dye-stained areas were 86.5–91.2% associated with sites having roots and were mainly distributed along the cracks in MC.

The actual penetration depth (H_a) and its frequency distribution demonstrated the variability of vertical preferential flow (Fig. 3). H_a in a single node (0.01 m × 0.01 m) was as deep as 0.55–0.65 m in the AF plots, 37–43% of

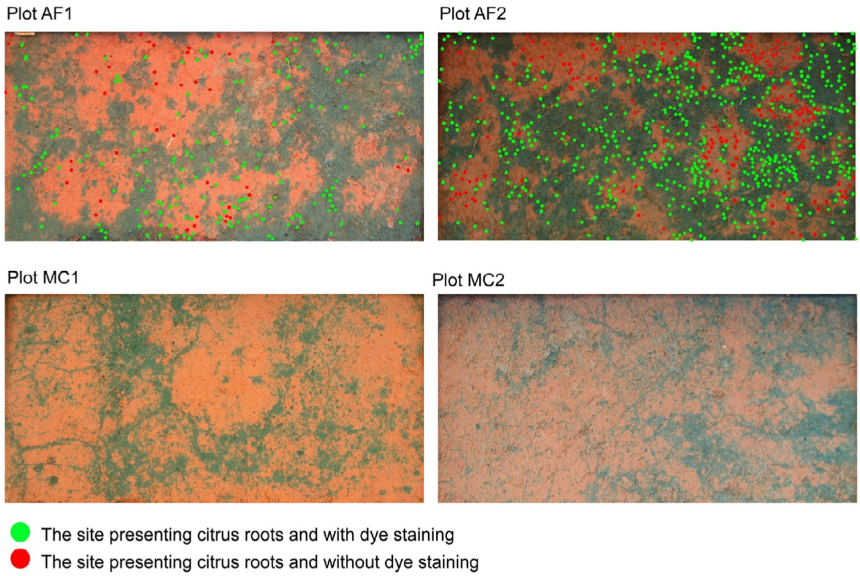

Fig. 2 Horizontal photographs showing dye-staining patterns at 0.10 m depth: within the frame (1.0 m × 1.0 m) in the agroforestry system (Plot AF1 and Plot AF2) and the monocropping system (Plot MC1 and Plot MC2) under the ponding infiltration without rainfall simulation. The *dots* in the photos of AF1 and AF2 indicated the sites presenting citrus roots, with the *green* showing stained dye and the *red* showing no stained dye.

which was distributed within the 0–0.10 m soil depth, while it was as shallow as 0.25–0.45 m in MC, 65–67% of which was concentrated within the 0–0.10 m soil depth. The mean H_a indicates the average percolation depth of vertical preferential flow in the metal frame. The mean H_a values were 0.13–0.17 m in AF, being deeper than 0.08 m in MC.

3.2 Horizontal Dye-Staining Patterns After a Simulated Rainfall

The horizontal photos taken after the rainfall simulation demonstrated that the dye-staining area extended out of the ring area upslope and downslope obviously at the 0.10 and 0.20 m soil depths in MC and only at 0.10 m soil depth in AF (Fig. 4). Differently, the dye staining diffused in a small domain around the steel ring at the 0.10 m soil depth in AF, but spread over the whole horizontal image at the 0.10 m and 0.20 m soil depth in MC. According to the dye coverage area change, the maximum distance of dye movement and the preferential flow contribution to

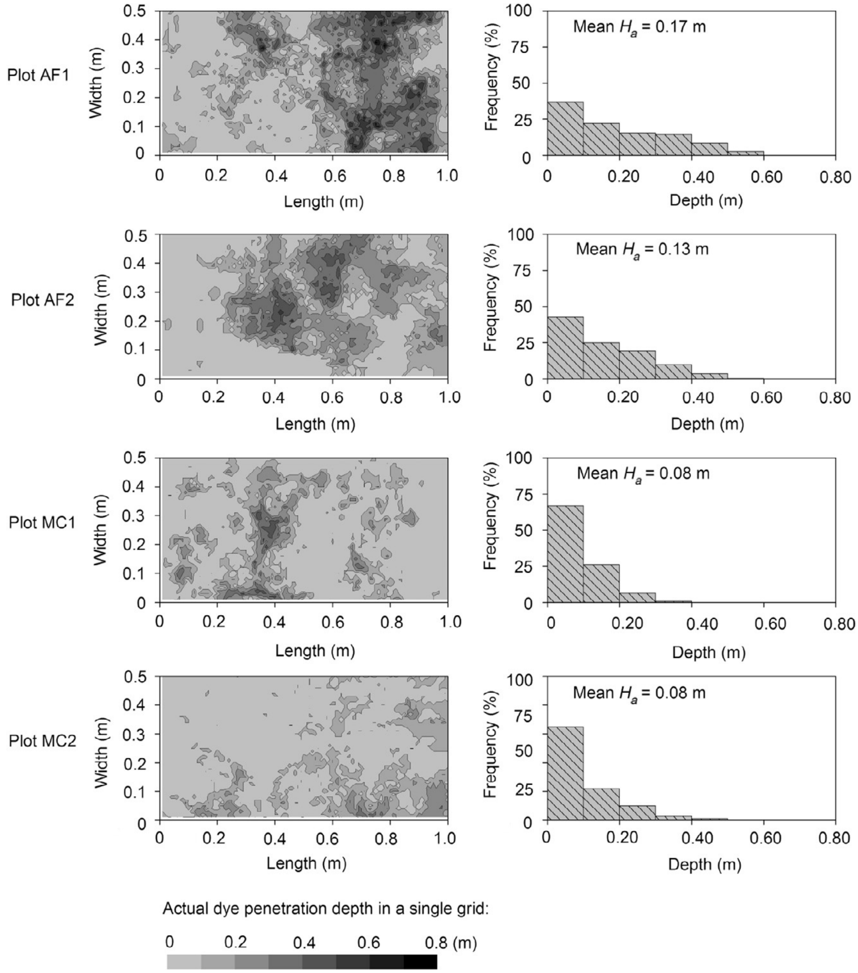

Fig. 3 Ordinary Kriging interpolation of the actual dye penetration depth in a single node (0.01 m × 0.01 m) (*left*) and its frequency distribution (*right*) based on horizontal photos from the dye-tracing experiment under ponding infiltration without rainfall simulation in the agroforestry system (Plot AF1 and AF2) and monocropping system (Plot MC1 and MC2). The *gray* intensity indicated the actual dye penetration depth in a single node (from 0 to 0.8 m), and mean H_a indicated the actual dye penetration depth at the whole plot scale (0.5 m width, 1.0 m length).

subsurface lateral flow were estimated (Table 3). The maximum movement distance was further downslope than upslope in the both systems and further in MC than in AF. The preferential flow contribution to subsurface lateral flow (C_P) was 21.3–54.6% in AF and 58.0–59.8% in MC.

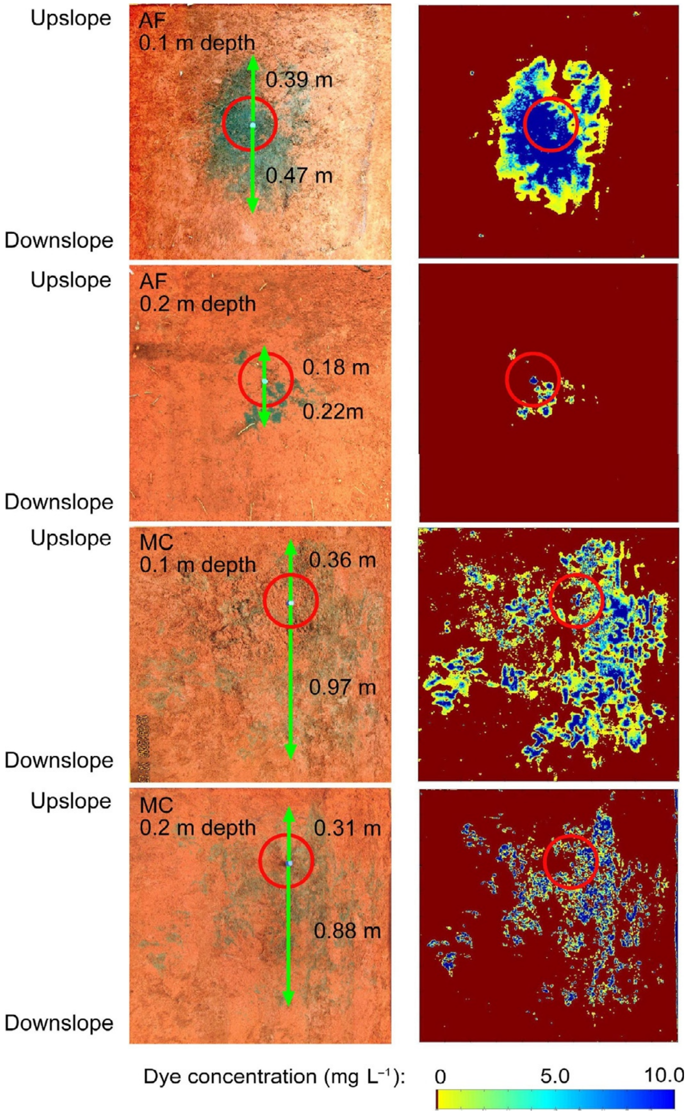

Fig. 4 Horizontal photographs and digitized images showing dye-strained areas surrounding the ring (*red circle*) at 0.1 and 0.2 m depth after rainfall simulation in the agroforestry (AF) and monocropping system (MC). *Different colors* in the images indicated a range of dye concentrations from 0 to 10.0 mg L^{-1}. The *green arrows* indicated the dye movement distance along upslope and downslope direction.

Table 3 Maximum Distance of Dye Movement (D_m) Upward and Downward, Dye Coverage (D_c), Dye Coverage of Preferential Flow ($D_{c\text{-}pf}$), Actual Distance of Dye Movement (D_{ad}), Preferential Flow Contribution to Subsurface Lateral Flow (C_p) in the Agroforestry (AF) and Monocropping System (MC) Applying the Dye-Tracing Experiment With Rainfall Simulation

Land Use	Depth (m)	D_m (m)		D_c (%)	$D_{c\text{-}pf}$[a] (%)	D_{ad}[b] (m)	C_p[c] (%)
		Upward	Downward				
Agroforestry	0.1	0.39	0.47	19.3	7.8	0.37	21.3
	0.2	0.18	0.22	1.5	0.3	0.10	54.6
Monocropping	0.1	0.36	0.97	21.6	8.5	0.39	59.8
	0.2	0.31	0.88	19.4	8.6	0.37	58.0

[a]Dye coverage of preferential flow estimated with dye concentration of >5 g L^{-1}.
[b]Actual distance of dye movement calculated using Eq. (3).
[c]Preferential flow contribution to subsurface lateral flow calculated using Eq. (2).

4. DISCUSSION

4.1 Agroforestry System Influences on Vertical Preferential Flow Pathways

The dye staining illustrates the hydrological pathways for soil water flow in the upper part of soil profiles in Earth's Critical Zone. The dye-staining patterns under no rainfall condition show that vertical preferential flow occurred in both cropping systems, but the vertical preferential flow pattern differed (Figs. 1 and 2). The vertical preferential flow likely developed more in the surface soil (above 0.15–0.17 m) but less in the subsoil (below 0.15–0.17 m) in MC than AF, which corresponded well to the plough pan in MC. This result is consistent with the results of Andreini and Steenhuis (1990), who observed the same phenomenon between no-tilled and tilled soils, which differed in the connectivity of macropores in the surface and subsoils. The horizontal images suggested that the hydrological pathways of vertical preferential flow in the plough layer were more related to identified roots, particularly senile or decaying roots in AF, and were more related to well-connected cracks in MC (Fig. 2). Cracks form due to wetting and drying cycles (Hendrickx and Flury, 2001; Zhang et al., 2014). The stained areas along the cracks were well associated with the presence of tree roots in AF, suggesting that the citrus roots developed vertical preferential flow in soils. Therefore, the interaction between soil structure and tree roots, particularly in deep soil, is the key triggering mechanism for vertical preferential flow in AF (Seobi et al., 2005; Weiler and

McDonnell, 2007). There were few roots that were associated with the stained cracks in the plough layer in MC (Fig. 2), demonstrating that tillage destroyed the root-generated macropores and made the soil more homogenous (Andreini and Steenhuis, 1990; Zhang et al., 2015b).

The actual dye penetration depth was deeper in AF than in MC (Fig. 3), indicating that the preferential flow pathways were better connected from the top to deeper soil in AF than in MC. Soil texture has been considered to be a major reason for spatial heterogeneity; however, it showed negligible differences between the contrasting land uses (Table 1). These results suggested that the distinguished dye penetration depths in AF and MC were not induced by different soil texture, but from different soil structure. Literature reported that plough pan in shallow-rooted agroecosystems may prevent growth of crop roots, resulting in a decrease in connectivity of macropores between the surface and subsurface soils if any (Bertolino et al., 2010; Verbist et al., 2007). Field excavation carried out in the field showed that 88% of peanut roots were distributed within the top 0.20 m soil layer, while 90% of the citrus roots concentrated within the top 0.85 m (Wang et al., 2011). These results suggested that tree roots may have penetrated through the plough pan in the peanut growth alley. Therefore, soil infiltration and vertical preferential flow increase when tree roots increase (Seobi et al., 2005; Weiler and McDonnell, 2007) or when the size and connectivity of macropores increase (Schaik, 2010; Trojan and Linden, 1992). This is also the reason why the actual dye penetration depth in a single node (0.01 m × 0.01 m) was mostly concentrated within the 0–0.10 m soil depth in MC (Fig. 3).

4.2 Agroforestry System Influences on Subsurface Lateral Flow

The interaction between vertical preferential flow and subsurface lateral flow determines the spatial variability of water flow and material transport in the upper part of soil profiles in Earth's Critical Zone. The dye-staining domain and the diffusion distance downslope were smaller in AF than in MC after the rainfall (Fig. 4), demonstrating less subsurface lateral flow in the plough layer in AF than in MC. This is confirmed by the smaller contribution of preferential flow to the subsurface lateral flow in AF than in MC (Table 3). The difference can be also attributed to the difference in the presence of plough pan between AF and MC. The absence of plough pan allowed free percolation of water toward deeper soils without flow resistance (Shaw et al., 2001). The plough pan, however, reduced the hydrological conductivity of the soils in the plough pan and reduced the hydrological connectivity between the surface and deep soils (Heppell et al., 2000; Horn and Smucker, 2005) and along the top and bottom positions along the slope transect. The well-connected cracks and identified decaying roots in the plough layer of MC (Fig. 2) likely facilitated the subsurface lateral flow through preferential flow along lateral roots (Schaik, 2010; Verbist et al., 2007). Therefore, the heavy rainfall may contribute more to subsurface lateral flow and less to vertical preferential flow in MC compared to AF. This is consistent with our previous study based on long-term monitoring in the same field site, which demonstrated that the subsurface lateral flow monitored during rainstorms was less, but the saturated water table was higher and more stable in AF than in MC (Wang et al., 2011). Here, it is confirmed that the subsurface lateral flow generated above the plough pan may be an important flow pathway in MC. When the plough layer is saturated in MC, overland flow may occur (Verbist et al., 2007). This agrees with higher overland flow in MC than in AF, also observed from erosion plots located in the same field (Tang et al., 2011).

Soil is the central interface of Earth's Critical Zone, representing a geomembrane across which water is actively exchanged with the atmosphere, biosphere, hydrosphere, and lithosphere (Lin, 2010). Once a small change occurred in soil surface, surface soil hydrological conditions and processes can dramatically vary in the upper soil profiles in Earth's Critical Zone. The traditional hydrology theory usually considers the pedosphere as a static porous medium, but neglects the dynamic influences of the biosphere on it (Lin, 2010). This is the reason why the function of AF systems to reduce subsurface lateral flow was largely attributed to increased evapotranspiration and enhanced canopy interception in previous studies (Dunin, 2002; Wang et al., 2011). This study suggested that the reduced subsurface lateral flow along the slope in AF can also be attributed to enhanced vertical connectivity and reduced lateral connectivity of hydrological pathways compared with MC, where tillage caused formation of a homogenous plough layer and compacted plough pan. Such small-scale soil heterogeneity will result in significant variability in soil flow pattern at a large scale (Allaire et al., 2009). However, this study provides qualitative results on the relationship of land-use practices to the interaction between vertical preferential flow and subsurface lateral flow as controlling hydrological processes of Earth's Critical Zone. Further studies are needed to quantify the relationship and controlling mechanisms as it is crucial for addressing major environmental and sustainability issues such as providing sustainable agricultural practices.

5. CONCLUSIONS

This study explored the land-use effects on vertical preferential flow and subsurface lateral flow as controlling hydrological processes of Earth's Critical Zone. The dye patterns before and after rainfall demonstrated that vertical preferential flow and subsurface lateral flow were generated in different pathways in the agroforestry and monocropping systems. The vertical preferential flow started from the surface soil to a deeper soil layer through pathways associated with tree root growth in the agroforestry system and below plough pan to a shallower layer through cracks in the monocropping system. The results suggest greater vertical preferential flow in the agroforestry system than in the monocropping system. During a rainfall event, the vertical preferential flow was redirected into near-surface lateral flow along the slope. This effect was greater through the plough layer in the monocropping system than in the agroforestry system. These differences in the hydrological pathways were due to different isotropy and heterogeneity of porous soil structure. These results suggest that land use has strong impacts on the distribution of water mass and flow paths not only overland but also in the subsurface, and that understanding the landscape hydrology in Earth's Critical Zone needs to simultaneously consider the coupled pedological and biological processes.

ACKNOWLEDGMENTS

This work was funded by the EC FP7 (SoilTrEC, Grant No. 244118), the Commonwealth Research Project of the Chinese Ministry of Water Resources (No. 201501055), and the Technology Research and Development Program of the Ministry of Science and Technology of China (No. 2014BAD14B01). Mr. Q.T. Zhang, Z.B. Zhang, and L. Lin are acknowledged for their contribution to the field work.

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